

160/230

Baro 2" 20'

42655
1 of 3

BENTELER Steel/Tube
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Tel: +1 318-216-4149
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BENTELER 
Steel/Tube

INSPECTION CERTIFICATE EN 10204-3.1
EN 10204:2005-01

Double Pipe Sales, Inc.
PO Box 300650
HOUSTON TX 77230
USA

Document No.: 455-1367038/001/P
Purchase Order No.: 11281
Benteler Order No.: 48002882
Dispatch Note No.:
Product: SEAMLESS STEEL TUBES

Inspection No.: Page: 1 / 3
Manufacturer: BSTMC Shreveport USA
Manufacturer's brand: 
Stamp of the inspection representative: M
Steelmaking process: ELECTRIC FURNACE

Terms of delivery: ASTM-A 53-2018, ASME SA-53, ASME BPVC.III-A-2021, ASTM-A 106-19a, ASME SA-106, ASME BPVC.III-A-2021, API Specification 5L
Forty-Sixth Edition April 2018 incl. Errata 1, May 2018, PSL 2, ANSINACE MRD175/ISO 15156-1: 2015, ANSINACE MRD175/ISO 15156-2:
2015, ANSINACE MRD103/ISO 17948:2015

Dimensions-tolerances: ASTM-A 106-19a, ASME SA-106, ASME BPVC.III-A-2021, ASTM-A 53-2018, ASME SA-53, ASME BPVC.III-A-2021, API Specification 5L
Forty-Sixth Edition April 2018 incl. Errata 1, May 2018

Steel grade: GRADE B, GRADE C, GRADE BN, GRADE X42N

Delivery condition: N

Product marking: FS: BENTELER BENTELER-Sign API 5L-1072 API-Sign Month/Year Conditions Dimension: O.D. * 1 inch Steel Grade
PSL2 SMLS Heat No. Tube length feet Test Pressure PSI Production No. Customer's material no.

ALZ = Etching ink marking FK = color marking FS = paint stenciling PSD = Color jet printer FSI = Color jet printer inside LK = Laser marking PVE = tag marking PS = die stamp TS = ink jet spray marking

Item	Dimensions	Heat No.	Test pressure PSI	sec
0004	2 WPS - SCH 80 35.000 FT - 45.000 FT	314786	2970	5

Country of melt and pour

Melted & Manufactured in USA
Bucor, Jewett, TX, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0004	314786	0.160	0.255	0.99	0.011	0.004	0.12	0.03	0.07	0.20	0.004	0.003	0.002	0.0003

- Formula: $CE \leq \frac{W}{100} = \frac{C}{100} + \frac{Mn}{5} + \frac{(Cr+Mo+V)}{5} + \frac{(Cu+Ni)}{15} \leq 0.43 \%$
- Formula: $Cr+Cu+Mo+Ni+V \leq 1.00 \%$
- Formula: $Nb+V \leq 0.06 \%$
- Formula: $Nb+Ti+V \leq 0.15 \%$

H20 230

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Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0004	314786	0.374	0.424	0.007	0.009

Product analysis [%]

Item	Heat No.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V	Nb	Ti	B
0004	314786	0.170	0.270	0.99	0.013	0.006	0.11	0.03	0.07	0.21	0.003	0.001	0.003	0.0003
0004	314786	0.170	0.270	0.98	0.012	0.006	0.11	0.03	0.08	0.20	0.003	0.001	0.003	0.0003

1. Formula: $CE = \frac{Wt\%C}{100} + \frac{(Mn\%)}{6} + \frac{((Cr\% + Mo\% + V\%))}{5} + \frac{(Cu\% + Ni\%)}{15} = 0.43\%$ 2. Formula: $Cr + Cu + Mo + Ni + V = 1.00\%$ 3. Formula: $Nb + V = 0.08\%$ 4. Formula: $Nb + Ti + V = 0.15\%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0004	314786	0.382	0.423	0.004	0.007
0004	314786	0.381	0.423	0.004	0.007

Test results

The tubes have been submitted to a leak tightness test by:	Hydrostatic test acc. to API 5L; Test pressure/time-record, the test records/charts shall be available for examination by the inspector	PASSED
The tubes are non destructive tested:	ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM-E 309; ET - Test drilled hole: 0.125 in.; ET - Notch shape/hole: drilled hole	PASSED
Visual inspection:	PASSED	Bending test: PASSED
		Dimensions examination: PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0.5/Rm
Requirements			IN	RT 0.5 % PSI 42100-71800	Rm PSI 70000-95000	AZ %		MAX 0.93
0004	000011	314786	0.738 X 0.224	51779	75710	34.00	19.00	0.68
0004	001562	314786	0.739 X 0.223	51053	76145	33.00	19.00	0.67